Physical Chemistry Cume April 17, 2009

Consider a one-dimensional real normalized wave function $\psi(x)$ where x is the x coordinate of the particle. Let us change this wave function to $\varphi(x) = N\psi(ax)$ where a is a constant. This is equivalent of stretching or compressing the wavefunction along the x axis, depending on whether a is smaller that 1 or greater than 1.

(a) What should N be for $\varphi(x)$ to remain normalized?

(b) If the kinetic energy associated with $\psi(x)$ is T, and a=0.5, what is the kinetic energy of $\varphi(x)$?

(c) The wavefunction for a harmonic oscillator is $N\exp[-(x/\alpha)^2]$ where α=10.76 pm , and N is a normalization constant. Make a freehand plot of this wavefunction, using N=1, for x=-30 pm to +30 pm. Notes: exp(y) is an alternative notation for for e^y; 1 pm=10^{-12} m. This corresponds to a harmonic oscillator with a mass of m=1 amu (atomic mass unit, about the mass of the H atom) and force constant k=500 J/m^2, about the hydrogen atom in the C-H bond.

(a) If the mass increases, for instance deuterium is substituted for H in the previous problem, the quantity α decreases. Indicate how this would change the wavefunction. Will it be wider or narrower? Will its maximum increase or decrease? (Proof is needed). All 25 p.

Help: If x=g(y), then $\int_a^b f(x)\,dx = \int_{g(a)}^{g(b)} f(g(y))\,(dx/dy)dy$ (substitution of the variable). Use this with y=ax.